MIL-C-39012/41A
19 April 1971
SUPERSEDING
MIL-C-39012/41
15 March 1969

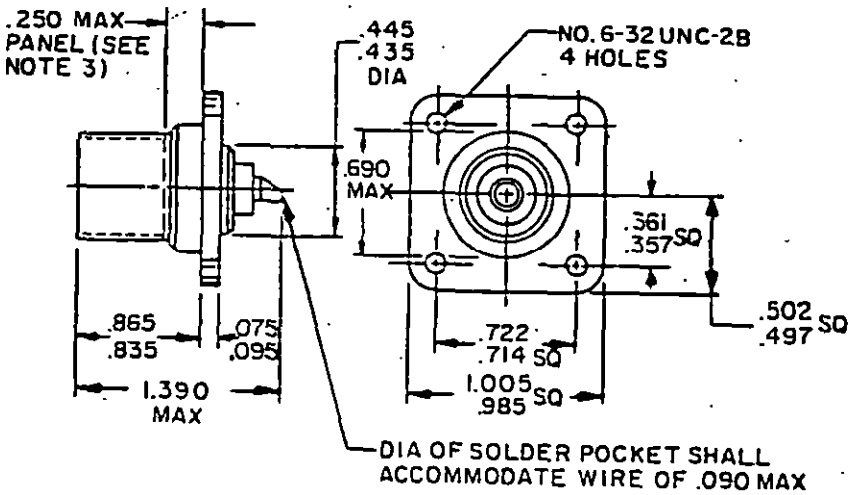
MILITARY SPECIFICATION SHEET

CONNECTORS, COAXIAL, RADIO FREQUENCY

SERIES SC (UNCABLED-RECEPTACLE, FEMALE. FLANGE, REAR MOUNTED, CLASS 2)

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-39012.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



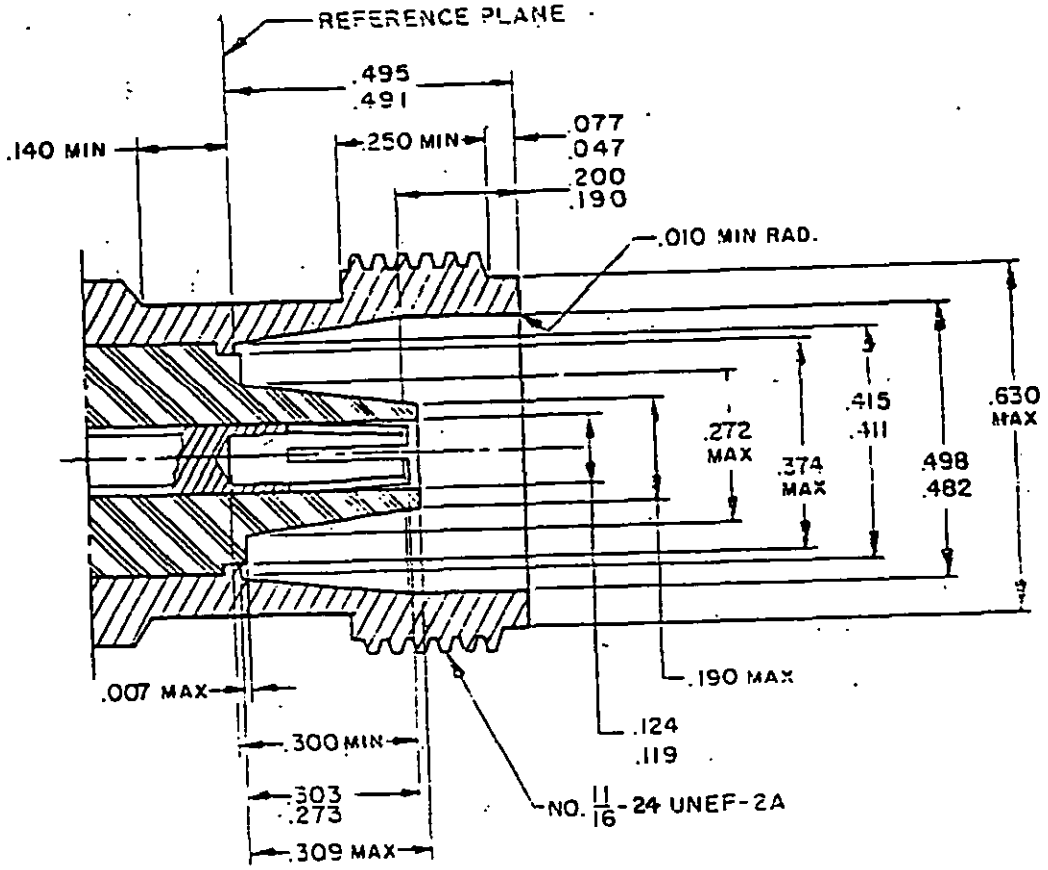
INCHES	MM	INCHES	MM
.075	1.91	.502	12.75
.090	2.29	.690	17.53
.095	2.41	.714	18.14
.250	6.35	.722	18.34
.357	9.07	.835	21.21
.361	9.17	.865	21.97
.435	11.05	.985	25.02
.445	11.30	1.005	25.53
.497	12.62	1.390	35.31

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Receptacle recommended for component part use and for use on panels with a .078 (1.98 MM) maximum thickness.
4. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 1. General configuration.

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:57Z
Check the source to verify that this is the current version before use.



INCHES	MM	INCHES	MM	INCHES	MM
.007	.18	.200	5.08	.411	10.44
.010	.25	.250	6.35	.415	10.54
.047	1.19	.272	6.91	.482	12.24
.077	1.96	.273	6.93	.491	12.47
.119	3.02	.300	7.62	.495	12.57
.124	3.15	.303	7.70	.498	12.65
.140	3.56	.309	7.85	.630	16.00
.190	4.83	.374	9.50		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 2. Mating dimensions for female terminations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:57Z
Check the source to verify that this is the current version before use.

ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating: 1,000 volts rms maximum working voltage at sea level, 250 volts rms maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:
Longitudinal force - Not applicable.
Torque - 6 inch pounds maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque - 6-10 inch pounds.

Mating characteristics:
See figure 2 for dimensions.
Center contact (female):
Oversize test pin - .098 diameter minimum, 16 microinch finish (non-closed entry contacts only).
Insertion depth - .094 minimum.
Number of insertions - 1.

Insertion force test - Steel test pin diameter .092 minimum.
Test pin finish - 16 microinches.
Insertion force - 2 lbs maximum.

Withdrawal force test: Steel test pin diameter .090 maximum.
Withdrawal force - 2 ounces minimum.
Test pin finish - 16 microinches.

Hermetic seal: Not applicable.

Leakage (pressurized connectors):
Not applicable.

Insulation resistance: Method 302, test condition B, MIL-STD-202. 5,000 megohms minimum.

Center contact retention: 6 lbs minimum axial force.

Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR):
Not applicable.

Swept frequency VSWR test setup:
Not applicable.

Connector durability:
500 cycles at 12 cycles/minute maximum. The connector shall meet mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact	1.0	1.5
Outer contact	.15	Not applicable
Braid to body	Not applicable	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 3,000 volts rms at sea level.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202.

Shock: Method 213 of MIL-STD-202, test condition I.

Temperature cycling: Method 102, test condition C, MIL-STD-202.

Thermal shock: Not applicable.

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:
Altitude - 70,000 feet.
Voltage - 750 volts rms minimum.

RF high potential withstanding voltage:
Voltage and frequency - 2,500 volts rms at 5 MHz.
Leakage current - Not applicable.

Cable retention force: Not applicable.

Coupling mechanism retention force:
Not applicable.

RF leakage: Not applicable.

Insertion loss: Not applicable.

Part number: 39012/41-0001.

Source: https://assist.dla.mil -- Downloaded: 202640-21T16:57Z
Check the source to verify that this is the currentversion before use.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI, MU, EL
Navy - OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - AT, AV, ME
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5935-1685-7)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:57Z
Check the source to verify that this is the current version before use.